OSSP Practical Experiments

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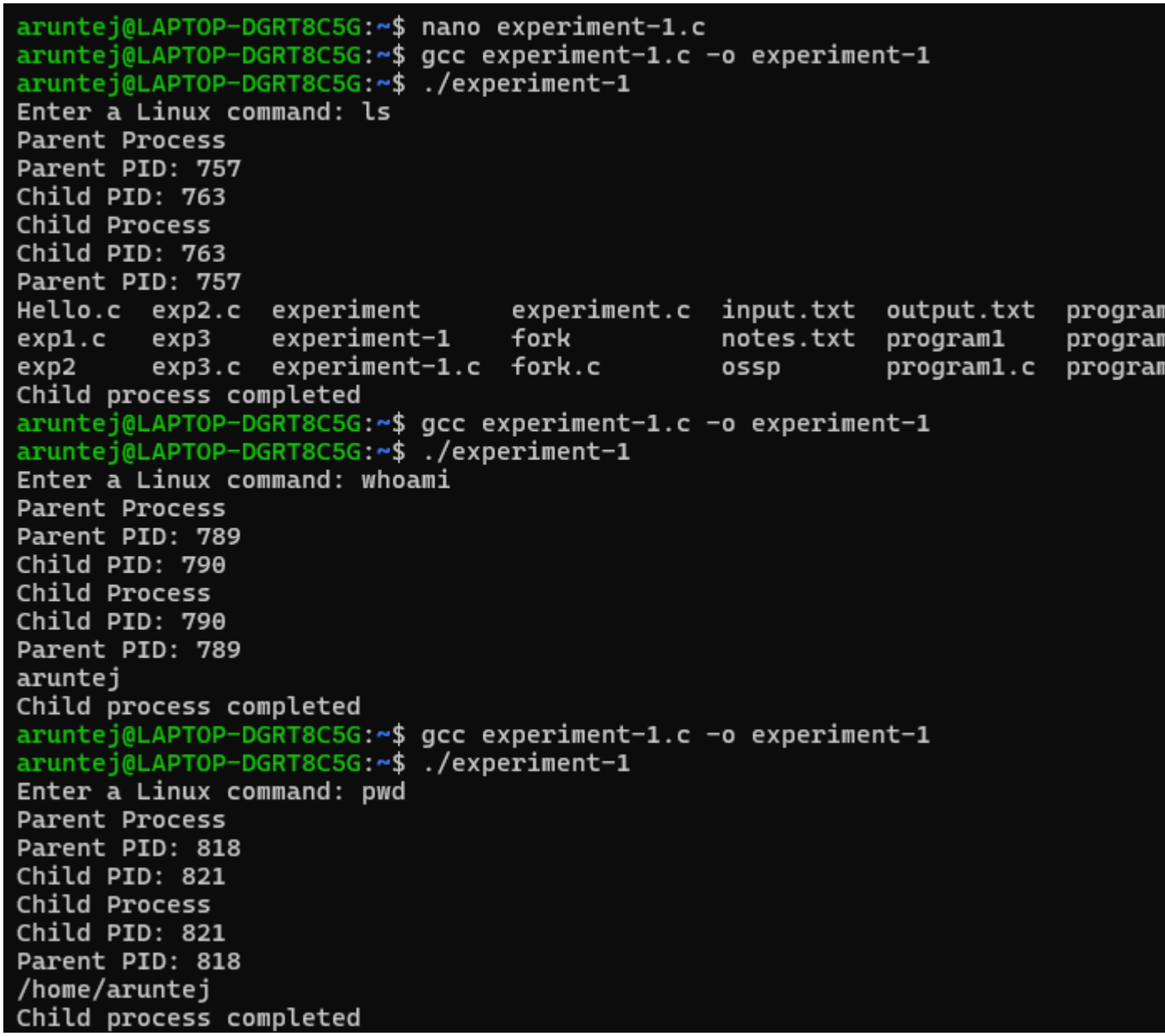
1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

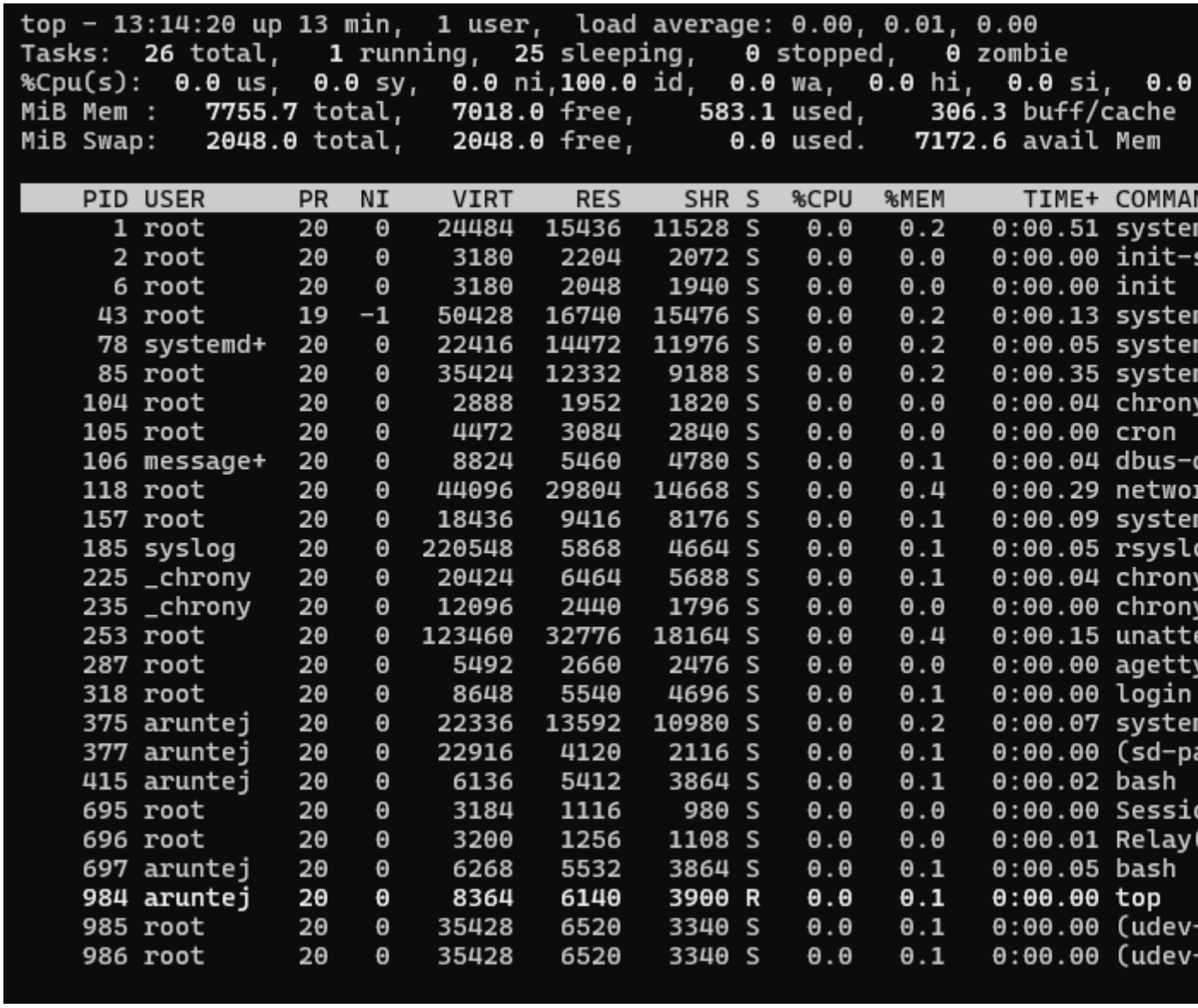
```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, NULL);

        printf("Command execution failed\n");
    }
    else
    {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed\n");
    }

    return 0;
}
```



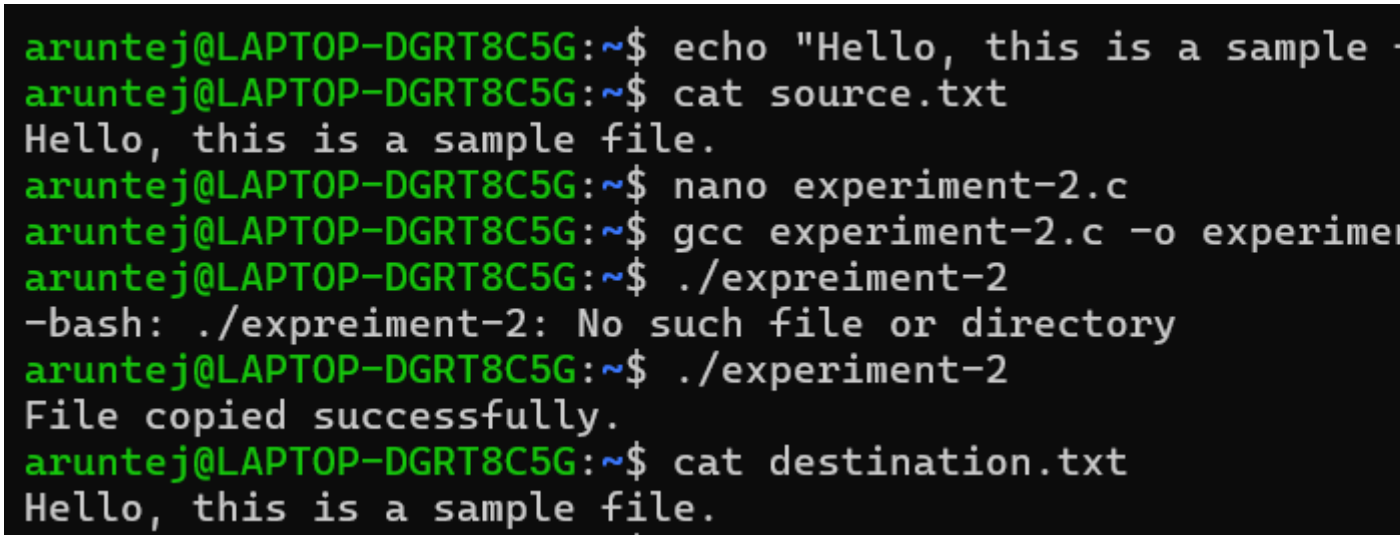
Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining

how the OS abstracts CPU, memory, storage, and I/O devices

```
aruntej@LAPTOP-DGRT8C5G:~$ nano experiment-1.c
aruntej@LAPTOP-DGRT8C5G:~$ gcc experiment-1.c -o experiment-1
aruntej@LAPTOP-DGRT8C5G:~$ ./experiment-1
Enter a Linux command: ls
Parent Process
Parent PID: 757
Child PID: 763
Child Process
Child PID: 763
Parent PID: 757
Hello.c  exp2.c  experiment      experiment.c  input.txt  output.txt  program
exp1.c   exp3    experiment-1    fork          notes.txt  program1    program
exp2     exp3.c  experiment-1.c  fork.c        ossp       program1.c  program
Child process completed
aruntej@LAPTOP-DGRT8C5G:~$ gcc experiment-1.c -o experiment-1
aruntej@LAPTOP-DGRT8C5G:~$ ./experiment-1
Enter a Linux command: whoami
Parent Process
Parent PID: 789
Child PID: 790
Child Process
Child PID: 790
Parent PID: 789
aruntej
Child process completed
aruntej@LAPTOP-DGRT8C5G:~$ gcc experiment-1.c -o experiment-1
aruntej@LAPTOP-DGRT8C5G:~$ ./experiment-1
Enter a Linux command: pwd
Parent Process
Parent PID: 818
Child PID: 821
Child Process
Child PID: 821
Parent PID: 818
/home/aruntej
Child process completed
```



2. Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.



Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.

```
aruntej@LAPTOP-DGRT8C5G:~$ echo "OSSP Practical Experiment - 2" > sample.txt
aruntej@LAPTOP-DGRT8C5G:~$ cat sample.txt
OSSP Practical Experiment - 2
aruntej@LAPTOP-DGRT8C5G:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7fffea0b9dc8 /* 25 vars */) =
brk(NULL)                                = 0x60d795bdc000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7
access("/etc/ld.so.preload", R_OK)       = -1 ENOENT (No such file or directory
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20763, ...}) = 0
mmap(NULL, 20763, PROT_READ, MAP_PRIVATE, 3, 0) = 0x71129210c000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEX
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 8
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7112920d900
mmap(0x7112920e0000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DE
mmap(0x711292101000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3,
mmap(0x711292108000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DEN
mmap(0x71129210a000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANO
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 8
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7112920ab00
mmap(0x7112920af000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DE
mmap(0x7112920d3000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3,
mmap(0x7112920d7000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DEN
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) =
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 8
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x711291f850
mmap(0x711291f97000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DE
mmap(0x711292037000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3
mmap(0x7112920a9000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DEN
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) =
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0\0\0"...,
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"...,
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"...,
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x711291c000
mmap(0x711291c28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_D
mmap(0x711291dc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3
mmap(0x711291e0e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DE
mmap(0x711291e14000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_AN
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEX
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 8
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x711291ed900
mmap(0x711291edc000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DE
mmap(0x711291f58000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3
mmap(0x711291f83000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DEN
```

```
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEX
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 8
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x711291ed900
mmap(0x711291edc000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DE
mmap(0x711291f58000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3
mmap(0x711291f83000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DEN
close(3)                                = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x
arch_prctl(ARCH_SET_FS, 0x711291ed4b00) = 0
set_tid_address(0x711291ed5128)         = 2072
set_robust_list(0x711291ed4de0, 24)     = 0
rseq(0x711291ed46a0, 0x20, 0, 0x53053053) = 0
mprotect(0x711291e0e000, 16384, PROT_READ) = 0
mprotect(0x711291f83000, 4096, PROT_READ) = 0
mprotect(0x7112920a9000, 4096, PROT_READ) = 0
mprotect(0x7112920d7000, 4096, PROT_READ) = 0
mprotect(0x711292108000, 4096, PROT_READ) = 0
mprotect(0x60d763563000, 1462272, PROT_READ) = 0
mprotect(0x711292157000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY
getrandom("\x31\xa2\x4d\x82\xd3\x6d\x8b\x6d", 8, GRND_NONBLOCK) = 8
munmap(0x71129210c000, 20763)           = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfr
SUID|ST_NODEV|ST_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7fff60124230)      = -1 ENOENT (No such file or directory
brk(NULL)                               = 0x60d795bdc000
brk(0x60d795bfd000)                     = 0x60d795bfd000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"..., 1024) = 442
lseek(3, 0, SEEK_CUR)                   = 442
lseek(3, 416, SEEK_SET)                 = 416
close(3)                                = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "/dev/sdd / ext4 rw,relatime,disc"..., 1024) = 1024
read(3, " tmpfs rw,nosuid,nodev,mode=755 "..., 1024) = 1024
read(3, ",upperdir=/gpu_lib/rw/upper,work"..., 1024) = 1024
read(3, "th=C:\\;uid=1000;gid=1000;symlink"..., 1024) = 299
read(3, "", 1024)                       = 0
close(3)                                = 0
access("/etc/selinux/config", F_OK)     = -1 ENOENT (No such file or directory
fcntl(0, F_GETFD)                       = 0
fcntl(1, F_GETFD)                       = 0
fcntl(2, F_GETFD)                       = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) =
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeou
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORE
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "60d762bfa000-60d762cec000 r--p 0"..., 1024) = 1024
read(3, "0 08:30 13647                   "..., 1024) = 1024
```

```
+++ exited with 0 +++
aruntej@LAPTOP-DGRT8C5G:~$ strace -e trace=openat,read,write,close cat sample
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOE
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 8
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXE
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 8
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) =
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 8
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) =
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0\0"...,
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOE
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 8
close(3)                                = 0
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"..., 1024) = 442
close(3)                                = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
read(3, "/dev/sdd / ext4 rw,relatime,disc"..., 1024) = 1024
read(3, " tmpfs rw,nosuid,nodev,mode=755 "..., 1024) = 1024
read(3, ",upperdir=/gpu_lib/rw/upper,work"..., 1024) = 1024
read(3, "th=C:\\;uid=1000;gid=1000;symlink"..., 1024) = 299
read(3, "", 1024)                       = 0
close(3)                                = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
read(3, "5639fd74f000-5639fd841000 r--p 0"..., 1024) = 1024
read(3, "0 08:30 13647                   "..., 1024) = 1024
read(3, "00 08:30 13650                  "..., 1024) = 1024
read(3, "             /usr/lib/x86_64-lin"..., 1024) = 1024
read(3, "0 13644                        /us"..., 1024) = 566
close(3)                                = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_
read(3, "# Common strings shared across a"..., 4080) = 4080
read(3, "", 32)                         = 0
close(3)                                = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLO
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
OSSP Practical Experiment - 2
close(5)                                = 0
close(4)                                = 0
close(3)                                = 0
+++ exited with 0 +++
aruntej@LAPTOP-DGRT8C5G:~$
```

3. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

   Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
GNU nano 6.7.1
int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("\n--- CHILD PROCESS ---\n");
        printf("PID  : %d\n", getpid());
        printf("PPID : %d\n", getppid());

        printf("Child is going to WAIT/SLEEP...\n");
        sleep(10);

        printf("Child is now RUNNING...\n");

        // CPU intensive work
        for (long long i = 0; i < 500000000; i++)
        {
            // Busy work
        }

        printf("Child is TERMINATING...\n");
        exit(0);
    }
    else
    {
        // Parent process
        printf("\n--- PARENT PROCESS ---\n");
        printf("PID       : %d\n", getpid());
        printf("PPID      : %d\n", getppid());
        printf("Child PID : %d\n", pid);

        printf("Parent is WAITING for child...\n");

        wait(NULL);

        printf("Child has TERMINATED.\n");
        printf("Parent is TERMINATING...\n");
    }

    return 0;
}
```

```
aruntej@LAPTOP-DGRT8C5G:~$ nano experiment-3.c
aruntej@LAPTOP-DGRT8C5G:~$ gcc experiment-3.c -o experiment-3
aruntej@LAPTOP-DGRT8C5G:~$ ./experiment-3

--- PARENT PROCESS ---
PID       : 2137
PPID      : 1774
Child PID : 2138
Parent is WAITING for child...

--- CHILD PROCESS ---
PID  : 2138
PPID : 2137
Child is going to WAIT/SLEEP...
Child is now RUNNING...
Child is TERMINATING...
Child has TERMINATED.
Parent is TERMINATING...
```

4. Write a C program where a parent process creates multiple child processes and synchronizes their completion using wait() and waitpid(). Compare the behavior of both functions.

```
aruntej@LAPTOP-DGRT8C5G:~$ nano experiment-4.c
aruntej@LAPTOP-DGRT8C5G:~$ gcc experiment-4.c -o experimer
aruntej@LAPTOP-DGRT8C5G:~$ ./experiment-4
Parent Process: PID = 2283
Child 1: PID = 2284, PPID = 2283

Parent using wait():
Child 2: PID = 2285, PPID = 2283
Child 3: PID = 2286, PPID = 2283
Child 1: PID = 2284 completed
wait(): Child PID 2284 terminated
Child 2: PID = 2285 completed
wait(): Child PID 2285 terminated

Parent using waitpid():
Child 3: PID = 2286 completed
waitpid(): Child PID 2286 terminated

All child processes completed.
aruntej@LAPTOP-DGRT8C5G:~$
```

Create a scenario where a child process becomes a zombie process. Investigate the process table and then modify the program to eliminate zombie processes using proper synchronization techniques

```
aruntej@LAPTOP-DGRT8C5G:~$ nano zombie.c
aruntej@LAPTOP-DGRT8C5G:~$ gcc zombie.c -o zombie
aruntej@LAPTOP-DGRT8C5G:~$ ./zombie
Parent: PID = 2325
Child PID = 2326
Parent is sleeping...
Child: PID = 2326
Child is terminating...
Parent exiting...
aruntej@LAPTOP-DGRT8C5G:~$
```

```
aruntej@LAPTOP-DGRT8C5G:~$ ps -ef | grep zombie
aruntej     2366    2331  0 16:17 pts/2    00:00:00 grep
aruntej@LAPTOP-DGRT8C5G:~$ nano zombie_fixed.c
aruntej@LAPTOP-DGRT8C5G:~$ gcc zombie_fixed.c -o zombie_f
aruntej@LAPTOP-DGRT8C5G:~$ ./zombie_fixed
Parent: PID = 2400
Child PID = 2401
Child: PID = 2401
Child is terminating...
Parent collected child process.
No zombie process remains.
aruntej@LAPTOP-DGRT8C5G:~$
```